Supplementary Materials for A Spatiotemporal Prior-Based Learning Framework for Zero-Shot Cross-Scale Path Planning

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I. Environment Modeling

The environment for most path planning is represented as a discrete grid map, where the workspace is divided into a finite set of cells. In accordance with this widely adopted formulation, the environment is modeled in the same manner in this work. Formally, the set of all grid nodes is defined as

$$\mathcal{V} = \{(x, y) \mid x = 1, \ldots, H,\ y = 1, \ldots, W\}, \tag{1}$$

where H and W denote the height and width of this map, respectively. Each cell (x, y) is associated with a binary occupancy indicator function:

$$\mathrm{occ}(x, y) \in \{0, 1\}, \tag{2}$$

where $\mathrm{occ}(x, y)$=1 indicates that the cell is occupied by an obstacle, and $\mathrm{occ}(x, y)$=0 denotes a free cell. Accordingly, the set of obstacle nodes is defined as

$$\mathcal{O} = \{(x, y) \in \mathcal{V} \mid \mathrm{occ}(x, y) = 1\}, \tag{3}$$

while the set of free nodes is given by:

$$\mathcal{F} = \mathcal{V} \setminus \mathcal{O}. \tag{4}$$

II. Details of experimental setting

A. Parameter Configuration for Model Training

The STAPNet module is optimized using the Adam optimizer with a initial learning rate of 1×10^{-3} and a weight decay $\lambda = 10^{-4}$. After approximately 6 hours of training in an 11×11 FoV setting, the STAPNet module achieved stable convergence.

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B. The paremeters of ACOs

For ACOs baselines, the hyperparameters as followed: the pheromone importance factor α was set to 0.1, the heuristic importance factor β was set to 3, and the pheromone evaporation rate was sampled from the interval $[0.1, 0.4)$. The pheromone reinforcement constant Q was set to 20.

C. Network Complexity Analysis

The proposed STAPNet comprises approximately $0.321{\times}10^6$ parameters ($\approx$1.28MB) and requires only $\approx$33.45MFLOPs per inference (for $W{=}H{=}11$). This lightweight design enables real-time deployment on embedded CPUs or low-power GPUs, in contrast to heavy vision backbones like ResNet-50 [1] or ViT [2].

III. Dataset Details

A diverse dataset is compiled by integrating maps from prior studies [3]–[5], covering diverse map sizes and obstacle layouts. This dataset is constructed by consolidating multiple public benchmarks, including the Various Obstacles Dataset [5], the Motion Planning Dataset [3], and the City/Street Map Dataset [6]. To further enrich scene diversity and capture realistic urban layouts, we additionally curated a collection of 50 city maps, which were converted into binary images with explicit obstacle annotations.

Based on this dataset, classical planners generate demonstrations on these maps with randomized planning instances, which are used to train STAPNet. TTo evaluate scalability, ImiPath is tested across five grid sizes: 11×11, 15×15, 20×20, 25×25, and 30×30, with all algorithms assessed on the same 100 randomly sampled instances for fair comparison.

IV. Ablation studies

A. Effect of Expert Demonstration Quality

To investigate how the capability of the expert planner influences the learning performance of ImiPath, a set of comparative experiments were conducted using demonstration data generated by different expert algorithms. **Table I** presents the performance of STAPNet trained with different experts under identical configuration settings.

The results clearly demonstrate that the quality of expert demonstrations directly influences the performance of the learned policy. When trained on trajectories generated by better expert planners, ImiPath achieves shorter average path lengths and higher completion rates. And it also demonstrates that higher-quality demonstrations provide more accurate guidance signals, enabling STAPNet to capture higher-quality decision patterns and generalize better to unseen environments. These findings underscore that expert-planner performance serves as a critical prior for data-driven planning and lays the groundwork for further prior-knowledge–guided planning strategy.

TABLE I
PERFORMANCE COMPARISON OF IMIPATH MODELS TRAINED WITH DEMONSTRATIONS GENERATED BY DIFFERENT EXPERT PLANNERS.

Method	*APL* $\pm$ *SD*	*Time(s)*	*CR(%)*
$\text{STAPNet}_{\text{AS}}$	5.551 ± 5.203	3.575×10^{-3}	94
$\text{STAPNet}_{\text{PFACO}}$	5.127 ± 3.951	3.338×10^{-3}	99

B. Effect of Dataset Size on Learning Performance

To examine how dataset scale affects the learning performance of ImiPath, five training sets comprising 1×10^4, 2×10^4, 1×10^5, 2×10^5 and 3×10^5 demonstrations, all generated by the same expert, were employed. The results in **Fig. 1 (a)** illustrate the convergence trends of STAPNet trained with different dataset sizes across training epochs. The vertical axis denotes the Win Ratio of ImiPath, and the horizontal axis denotes successive training epochs. As shown in Fig. 1 (a), larger datasets provide more diverse and informative state–action pairs, facilitating faster convergence and more stable policy learning. When the dataset size increases to the range of approximately 2×10^5 to 3×10^5, the performance curves begin to plateau, and the differences across dataset sizes become small, suggesting a clear trend of diminishing returns from additional data.

These results highlight that ImiPath achieves reliable convergence and strong performance even when trained with 2×10^5 scale datasets. This demonstrates the efficiency of the proposed learning architecture and its ability to extract effective decision patterns from limited-size demonstrations, further validating the practical applicability of ImiPath in data-constrained scenarios.

C. Effect of Different Network structures on Learning Performance

This section evaluates how STAPNet architectural modules influence learning. Six variants are trained under identical settings: MLP, PolicyHead (Only), CNN + PolicyHead, FCN + PolicyHead, STAPNet (Spatiotemporal Encoder + PolicyHead), and the end-to-end ImiPath system. **Fig. 1 (b)** illustrates the Win Ratio across training epochs for different network architectures trained under identical configurations. The results show that STAPNet attains the highest steady-state Win Ratio among all compared architectures and reaches its levels off

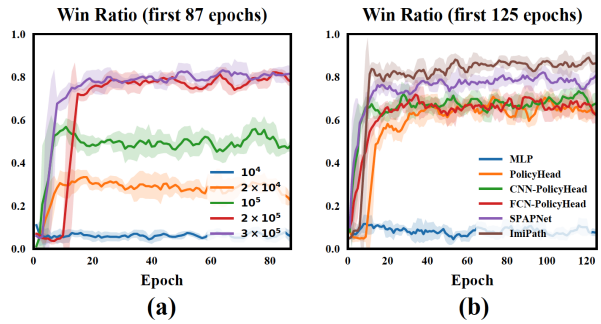

Fig. 1. Effects of dataset scale and STAPNet architecture on decision-making performance. (a) shows the decision-making performances for five training sets at different scales; (b) shows the decision-making performances for multiple STAPNet architectural variants and the end-to-end ImiPath system.Shaded areas show the 95% confidence interval.

rapidly, with narrow confidence bands across runs, which is consistent with faster convergence and stable decision quality. PolicyHead converges quickly but stabilizes at a lower level, suggesting that a head alone provides insufficient capacity to encode spatiotemporal information. CNN + PolicyHead and FCN + PolicyHead achieve intermediate performance with slower convergence and larger variability, indicating partial capture of spatial structure or temporal features but limited integration of spatial and temporal information. MLP underperforms and shows the greatest instability, implying that non-convolutional, non-temporal features are inadequate for this task. Finally, ImiPath, which integrates STAPNet within the overall planning pipeline, attains the highest curve with consistently narrow variability, indicating that the complete framework further improves decision quality and stability beyond the encoder–head module alone.

The results indicate that module design substantially affects decision-making performance. Combining a compact policy Head with an encoder that integrates spatial and temporal information yields more stable, higher-quality decisions and faster convergence. Furthermore, incorporating this module into the full ImiPath framework yields enhanced improvements in performance and stability.

TABLE II
ABLATION RESULTS OF DIFFERENT HYBRID STRATEGIES FOR DETERMINISTIC-BASED IMIPATH. COMPARISON AMONG THE IMIPATH AND TWO FUSION STRATEGIES THAT INTEGRATE STAPNET PREDICTIONS INTO THE DETERMINISTIC PLANNER.

Mehtod[1]	*APL* $\pm$ *SD*	*Time(s)*	*CR(%)*	APL_p
$\text{STAPNet}_{\text{A}*}$	5.283 ± 4.947	3.23×10^{-3}	94	$\times$
$\text{ImiPath}_{(f_1)}$	5.695 ± 3.736	1.172×10^{-2}	100	–
$\text{ImiPath}_{(f_2))}$	5.001 ± 3.369	1.91×10^{-2}	100	–
$\text{ImiPath}_{(f_3)}$	4.868 ± 3.268	1.09×10^{-2}	100	

[1] g denotes the accumulated path cost from the start to the current node, and h denotes the heuristic estimate of the remaining cost to the goal in A*. φ denotes the selection probability of the feasible actions predicted by STAPNet

TABLE III
ABLATION RESULTS OF DIFFERENT HYBRID STRATEGIES FOR STOCHASTIC-BASED IMIPATH. COMPARISON AMONG IMIPATH AND TWO HYBRID VARIANTS (V1 AND V2) THAT INTEGRATE STAPNET PREDICTIONS INTO THE STOCHASTIC PLANNER.

	APL $\pm$ *SD*	*Time(s)*	*CR(%)*	APL_p
NO Hybrid				
ImiPath-30-20$_{\text{AS}}$	5.551 ± 5.203	9.529×10^{-2}	94	$\times$
ImiPath-30-20$_{\text{PFACO}}$	5.127 ± 3.951	7.941e-01	100	$\sim$
Hybrid v1				
ImiPath-30-20$_{\text{AS}}$	5.216 ± 3.782	1.461	100	–
ImiPath-30-20$_{\text{PFACO}}$	4.912 ± 3.340	9.841×10^{-1}	100	–
Hybrid v2				
ImiPath-30-20$_{\text{AS}}$	5.175 ± 3.785	1.341	100	–
ImiPath-30-20$_{\text{PFACO}}$	4.767 ± 3.088	3.369×10^{-1}	100	

D. Effect of Different Prior Knowledge-Guided Planning Strategy

To further examine the effectiveness of the proposed prior knowledge-guided planning strategy, comparative experiments were conducted using different expert planners and policy fusion strategies. **Table II** and **Table III** summarize the results for the determistic-based and stochastic-based variants of ImiPath, respectively. For the deterministic A*-based paradigm (as shown in Table II), STAPNet priors were embedded into the node evaluation rule. Three formulations were examined, namely $f_1 = g+\varphi$, $f_2 = g+h+(1-\varphi)$ and $f_2 = g+(1-\varphi)$. ImiPath$_{(f_3)}$ produced the shortest average path length and achieved a 100% completion rate, confirming that incorporating STAPNet predictions effectively refines the heuristic estimation and accelerates convergence. For the stochastic ACO-based paradigm (as shown in Table III), two hybrid strategies were designed. Hybrid v1 initializes the pheromone concentration matrix using STAPNet predictions before the search begins, whereas Hybrid v2 integrates the predicted policy into every iteration to continuously guide action selection. The results show that Hybrid v2 consistently outperforms both the NO Hybrid and Hybrid v1, demonstrating that iterative policy fusion yields more efficient and stable search behavior than one-time initialization.

These hybrid experiments demonstrate that continuous policy guidance can effectively enhance both stochastic and deterministic planning approaches. By iteratively incorporating predictions from STAPNet, planners are able to dynamically balance exploration and exploitation, leading to improved convergence efficiency and higher-quality solutions. These findings validate the effectiveness of the proposed prior-guided fusion mechanism within the ImiPath framework.

V. DETAILS OF PHYSICAL EXPERIMENTS

Motivated by the stringent real-time decision-making requirements of minimally invasive microrobotic navigation through narrow channel-like structures under locally observable environments, the framework is instantiated on a magnetic microrobot platform [7].

The platform comprises three main components: a magnetic field generation and control module, a visual feedback system, and an upper-level motion planning interface. Motivated by applications in minimally invasive procedures and targeted therapy, the experimental environment emulates navigation challenges found in vascular or microchannel networks. A channel-type map is constructed within the magnetic workspace to reproduce narrow and tortuous pathways similar to human vasculature. This setup enables direct assessment of the local observation–based planning strategy in confined spaces that require boundary avoidance and adaptation to complex geometric constraints. By replicating microvascular structures in a controlled laboratory environment, the system provides a realistic and repeatable testbed for evaluating the ability of ImiPath to achieve safe, precise, and adaptive microrobot navigation in medically relevant scenarios.

The experimental channel map measures 76 mm × 76 mm and is discretized into a grid representation. ImiPath operates under local observations, where the robot's FoV is limited to a 11 mm × 11 mm. This constraint emulates the restricted sensing capabilities of microrobots operating in vivo, ensuring that navigation decisions rely solely on the currently observed area. As the robot moves, the local map is incrementally updated, enabling the ImiPath to adapt to new environmental information in real time. It is implemented on the upper-level computer, which inferences promising actions and transmits the corresponding control signals to the magnetic field generator through the robust backstepping sliding mode control framework. This setup ensures accurate trajectory tracking within the narrow, tortuous pathways of the channel environment.

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